

RESEARCH INTERESTS	Using photorealistic simulation for embodied AI and computer vision ; motion planning, trajectory optimization, and control methods for robotics; reconstructing 3D scenes from images; continuous and discrete optimization; software tools and algorithms for creativity support	
EDUCATION	Stanford University	Stanford, California
	Ph.D. Computer Science	2012–2019
	<i>Advisor: Pat Hanrahan</i>	
	<i>Dissertation: Trajectory Optimization Methods for Drone Cameras</i>	
	Harvard University	Cambridge, Massachusetts
	Visiting Research Fellow	Summer 2013
	<i>Advisor: Hanspeter Pfister</i>	
	University of Calgary	Calgary, Canada
	M.S. Computer Science	2010
	University of Calgary	Calgary, Canada
	B.S. Computer Science	2007
EMPLOYMENT	Adobe Research	San Francisco, California
	Senior Research Scientist	2024–
	<i>Mentor: Kalyan Sunkavalli</i>	
	Intel Labs	Seattle, Washington
	Research Scientist	2021–2024
	<i>Mentor: Vladlen Koltun</i>	
	Apple	Seattle, Washington
	Research Scientist	2018–2021
	<i>Mentor: Josh Suskind</i>	
	Microsoft Research	Redmond, Washington
	Research Intern	Summer 2016, Summer 2017
	<i>Mentors: Neel Joshi, Sudipta Sinha</i>	
HONORS & AWARDS	Skydio	Redwood City, California
	Research Intern	Spring 2016
	<i>Mentors: Adam Bry, Frank Dellaert</i>	
	Udacity	Mountain View, California
	Course Developer, Introduction to Parallel Computing	2012–2013
	<i>Instructors: John Owens, David Luebke</i>	
	Harvard University	Cambridge, Massachusetts
	Research Fellow	2010–2012
	<i>Advisor: Hanspeter Pfister</i>	
	NVIDIA	Austin, Texas
	Developer Tools Programmer Intern	Summer 2009
	Radical Entertainment	Vancouver, Canada
	Graphics Programmer Intern	2005–2006
	Featured in the Highlights of SIGGRAPH session at the FMX Festival 2017	2017
	<i>1% selection rate (3 / 467)</i>	
	Invited speaker, TEDxBerkeley 2017	2017
	Excellent reviewer, ACM Human Factors in Computing Systems (CHI) 2017	2017
	Featured in the SIGGRAPH 2016 Technical Papers Trailer	2016
	<i>4% selection rate (19 / 467)</i>	

Featured in the SIGGRAPH Asia 2015 Technical Papers Trailer <i>4% selection rate (11 / 302)</i>	2015
Front cover article, Cell 162(3)	2015
NSERC Alexander Graham Bell Canada Graduate Scholarship <i>3-year fellowship for Canadian PhD students, 14% selection rate (233 / 1628)</i>	2012

SOFTWARE & DATASETS

SPEAR: A Simulator for Photorealistic Embodied AI Research
github.com/spear-sim/spear

Hypersim: A Photorealistic Synthetic Dataset for Holistic Indoor Scene Understanding
github.com/apple/ml-hypersim

SELECTED PUBLICATIONS

My publications are also listed on [Google Scholar](https://scholar.google.com/citations?user=8333333333333333).

SPEAR: A Simulator for Photorealistic Embodied AI Research

Mike Roberts, Renhan Wang, Rushikesh Zavar, Rachith Dey-Prakash, Quentin Leboutet, Stephan R. Richter, Matthias Müller, German Ros, Rui Tang, Stefan Leutenegger, Yannick Hold-Geoffroy, Kalyan Sunkavalli, Vladlen Koltun
European Conference on Computer Vision (ECCV) 2026

PISA Experiments: Exploring Physics Post-Training for Video Diffusion Models by Watching Stuff Drop
 Chenyu Li, Oscar Michel, Xichen Pan, Sainan Liu, **Mike Roberts**, Saining Xie
International Conference on Machine Learning (ICML) 2025

Hypersim: A Photorealistic Synthetic Dataset for Holistic Indoor Scene Understanding

Mike Roberts, Jason Ramapuram, Anurag Ranjan, Atulit Kumar, Miguel Angel Bautista, Nathan Paczan, Russ Webb, Joshua M. Susskind
International Conference on Computer Vision (ICCV) 2021

Submodular Trajectory Optimization for Aerial 3D Scanning

Mike Roberts, Debadepta Dey, Anh Truong, Sudipta Sinha, Shital Shah, Ashish Kapoor, Pat Hanrahan, Neel Joshi
International Conference on Computer Vision (ICCV) 2017

Generating Dynamically Feasible Trajectories for Quadrotor Cameras

Mike Roberts, Pat Hanrahan
ACM Transactions on Graphics 35(4) (SIGGRAPH 2016)
Featured in the Highlights of SIGGRAPH session at the FMX Festival 2017
Featured in the SIGGRAPH 2016 Technical Papers Trailer

An Interactive Tool for Designing Quadrotor Camera Shots

Niels Joubert*, **Mike Roberts***, Anh Truong, Floraine Berthouzoz, Pat Hanrahan
ACM Transactions on Graphics 34(6) (SIGGRAPH Asia 2015), * Authors contributed equally
Featured in the SIGGRAPH Asia 2015 Technical Papers Trailer

Saturated Reconstruction of a Volume of Neocortex

Narayanan Kasthuri, Kenneth Jeffrey Hayworth, Daniel Raimund Berger, Richard Lee Schalek, Jose Angel Conchello, Seymour Knowles-Barley, Dongil Lee, Amelio Vazquez-Reina, Verena Kaynig, Thouis Raymond Jones, **Mike Roberts**, Josh Lyskowski Morgan, Juan Carlos Tapia, H. Sebastian Seung, William Gray Roncal, Joshua Tzvi Vogelstein, Randal Burns, Daniel Lewis Sussman, Carey Eldin Priebe, Hanspeter Pfister, Jeff William Lichtman
Cell 162(3), 2015
Front cover article

Large-Scale Automatic Reconstruction of Neuronal Processes from Electron Microscopy Images

Verena Kaynig, Amelio Vazquez-Reina, Seymour Knowles-Barley, **Mike Roberts**, Thouis R. Jones, Narayanan Kasthuri, Eric Miller, Jeff Lichtman, Hanspeter Pfister
Medical Image Analysis 22(1), 2015

Design and Evaluation of Interactive Proofreading Tools for Connectomics

Daniel Haehn, Seymour Knowles-Barley, **Mike Roberts**, Johanna Beyer, Narayanan Kasthuri, Jeff W. Lichtman, Hanspeter Pfister
IEEE Transactions on Visualization and Computer Graphics 20(12) (SciVis 2014)

Neural Process Reconstruction from Sparse User Scribbles

Mike Roberts, Won-Ki Jeong, Amelio Vazquez-Reina, Markus Unger, Horst Bischof, Jeff Lichtman, Hanspeter Pfister
Medical Image Computing and Computer Assisted Intervention (MICCAI) 2011

A Work-Efficient GPU Algorithm for Level Set Segmentation

Mike Roberts, Jeff Packer, Mario Costa Sousa, Joseph Ross Mitchell
High Performance Graphics 2010

**INVITED
TALKS****Sample-Efficient Learning with Synthetic Data**

Imperial College London	February 2025
University College London	
Boston Dynamics AI Institute	August 2024
Adobe Research	July 2024
Manycore Tech	September 2023
Toyota Research Institute	April 2021
Facebook AI Research	January 2021
Stanford University	October 2020
Intel Labs	
University of Washington	

Trajectory Optimization Methods for Drone Cameras

Oculus Research	June 2018
Snapchat Research	May 2018
Carnegie Mellon University	
Boston University	March 2018
Google Research	
Adobe Research	
Toyota Technological Institute at Chicago	
NVIDIA Research	February 2018
Simon Fraser University	

Harnessing the Creative Power of Drones

Charles University in Prague	November 2017
Google	June 2017
University College London	May 2017
Disney Research	
ETH Zurich	
University of Oxford	
Max Planck Institute for Informatics	
UC Berkeley	April 2017
Samsung	
TEDxBerkeley 2017	
Autel Robotics	March 2017
3D Robotics	
UC Berkeley	February 2017
Columbia University	November 2016
Yale University	
Princeton University	

	Brown University	
	Intel	October 2016
	Generating Dynamically Feasible Trajectories for Quadrotor Cameras	
	FMX Festival 2017, Highlights of SIGGRAPH session	May 2017
	Adobe Research	September 2016
	Apple	August 2016
	Massachusetts Institute of Technology	
	Skydio	February 2016
	3D Robotics	January 2016
TEACHING EXPERIENCE	Udacity	2013–2018
	Course Developer, Introduction to Parallel Programming	
	<i>Instructors: John Owens, David Luebke</i>	
	<i>Developed course materials in 2012–2013, over 80,000 students enrolled in 2013–2018</i>	
	Stanford University	Spring 2018
	Course Assistant, Convolutional Neural Networks for Visual Recognition	
	<i>Instructors: Fei-Fei Li, Justin Johnson, Serena Yeung</i>	
	Stanford University	Winter 2018
	Course Assistant, Mathematical Methods for Robotics, Vision, and Graphics	
	<i>Instructor: Doug James</i>	
	Massachusetts Institute of Technology	Summer 2016
	Guest Lecturer, Advances in Imaging	
	<i>Instructor: Ramesh Raskar</i>	
	Harvard University	Fall 2013
	Course Contributor, Data Science	
	<i>Instructors: Hanspeter Pfister, Joe Blitzstein</i>	
	<i>Contributed lecture notes to the initial offering of Harvard's Data Science course in Fall 2013</i>	
	Harvard University	Winter 2012
	Teaching Fellow, Visualization	
	<i>Instructor: Hanspeter Pfister</i>	
	Harvard University	Fall 2011
	Teaching Fellow, Computing Foundations for Computational Science	
	<i>Instructor: Hanspeter Pfister</i>	
	Harvard University	Winter 2011
	Teaching Fellow, Massively Parallel Computing	
	<i>Instructors: Hanspeter Pfister, Nicolas Pinto</i>	
	University of Calgary	Winter 2006, 2007, 2008
	Guest Lecturer, Video Game Programming	
REVIEWING EXPERIENCE	Conference	
	Computer Vision and Pattern Recognition (CVPR); Eurographics; International Conference on 3D Vision (3DV); International Conference on Robotics and Automation (ICRA); High Performance Graphics (HPG); Human Factors in Computing Systems (CHI); Neural Information Processing Systems (NeurIPS); SIGGRAPH; SIGGRAPH Asia	
	Journal	
	Transactions on Graphics (TOG); Robotics and Automation Letters (R-AL); Transactions on Visualization and Computer Graphics (TVCG)	
GAME CREDITS	Scarface: The World Is Yours (PC, Playstation 2, Wii, Xbox)	2006
	<i>Radical Entertainment, Sierra</i>	

PRESS COVERAGE

New App Lets Drone Pilots Customize Flight Path and Camera Movement Before Takeoff
Digital Trends (October 19th, 2015)

Researchers Create Software for Designing Pro Drone Shots in a Virtual World
Petapixel (October 16th, 2015)

Interactive Drone App Lets You Capture Aerial Shots Like a Pro
Engadget (October 15th, 2015)

These Stunning Images Will Take You on a Journey Through the Brain
Huffington Post (August 4th, 2015)

3D Color Images of the Brain Reveal its Glorious Unseen Detail
Popular Science (July 31st, 2015)

3D Brain Map Reveals Connections Between Cells in Nano-Scale
The Guardian (July 30, 2015)

Crumb of Mouse Brain Reconstructed in Full Detail
Nature News (July 30, 2015)

A Voyage into the Brain
National Geographic (February 2014)

What Makes Us Human?
BBC Horizon (July 3rd, 2013)

In Pursuit of a Mind Map, Slice by Slice
The New York Times (December 27th, 2010)

REFERENCES

Pat Hanrahan
CANON USA Professor of Computer Science and Electrical Engineering, Stanford University

Vladlen Koltun
Distinguished Scientist, Apple

Hanspeter Pfister
An Wang Professor of Computer Science, Harvard University

Adam Finkelstein
Professor of Computer Science, Princeton University

John Owens
Child Family Professor of Engineering and Entrepreneurship, UC Davis

Sudipta Sinha
Principal Researcher, Microsoft Research

Josh Susskind
Machine Learning Scientist, Apple